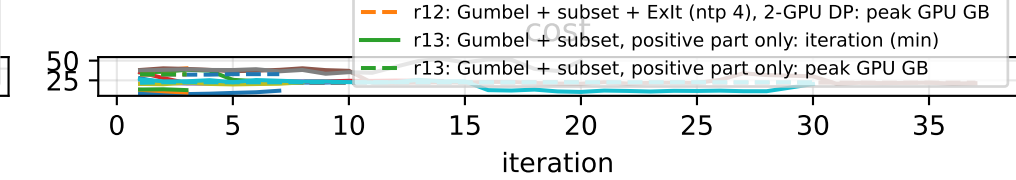
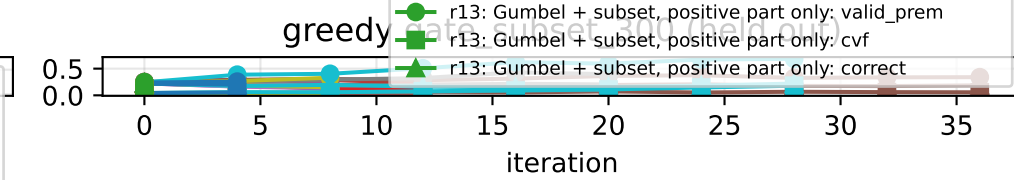
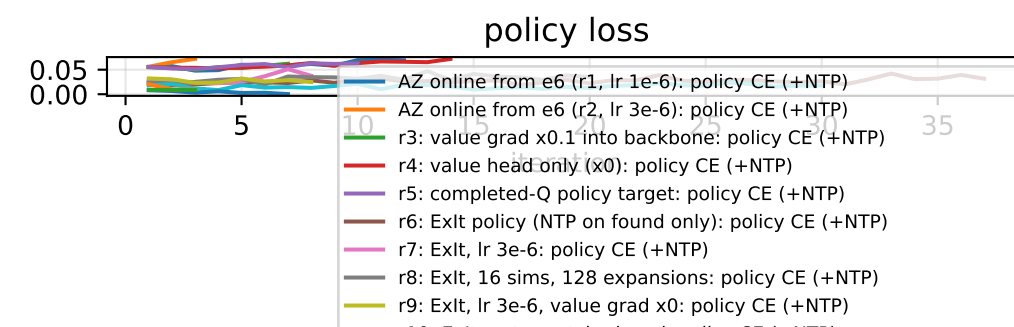
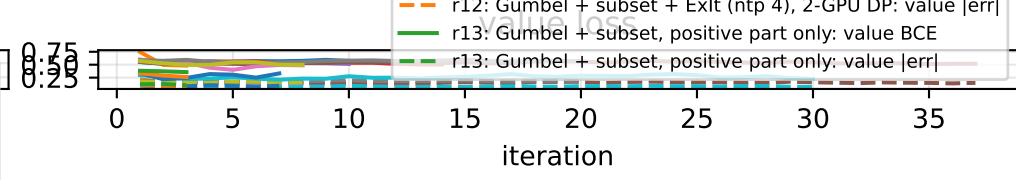
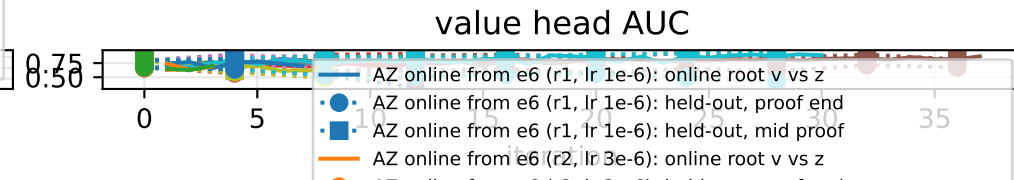


Online AlphaZero: policy + value head trained jointly from MCTS self-play



- AZ online from e6 (r1, lr 1e-6): online root v vs z
- AZ online from e6 (r1, lr 1e-6): held-out, proof end
- AZ online from e6 (r1, lr 1e-6): held-out, mid proof
- AZ online from e6 (r2, lr 3e-6): online root v vs z
- AZ online from e6 (r2, lr 3e-6): held-out, proof end
- AZ online from e6 (r2, lr 3e-6): held-out, mid proof
- AZ online from e6 (r1, lr 1e-6): valid_prem
- AZ online from e6 (r1, lr 1e-6): cvf
- AZ online from e6 (r1, lr 1e-6): correct
- AZ online from e6 (r2, lr 3e-6): valid_prem
- AZ online from e6 (r2, lr 3e-6): cvf
- AZ online from e6 (r2, lr 3e-6): correct
- r3: value grad x0.1 into backbone: valid_prem
- r3: value grad x0.1 into backbone: cvf
- r3: value grad x0.1 into backbone: correct
- r4: value head only (x0): valid_prem
- r4: value head only (x0): cvf
- r4: value head only (x0): correct
- r5: completed-Q policy target: valid_prem
- r5: completed-Q policy target: cvf
- r5: completed-Q policy target: correct
- r6: Exit policy (NTP on found only): valid_prem
- r6: Exit policy (NTP on found only): cvf
- r6: Exit policy (NTP on found only): correct
- r7: Exit, lr 3e-6: valid_prem
- r7: Exit, lr 3e-6: cvf
- r7: Exit, lr 3e-6: correct
- r8: Exit, 16 sims, 128 expansions: valid_prem
- r8: Exit, 16 sims, 128 expansions: cvf
- r8: Exit, 16 sims, 128 expansions: correct
- r9: Exit, lr 3e-6, value grad x0: valid_prem
- r9: Exit, lr 3e-6, value grad x0: cvf
- r9: Exit, lr 3e-6, value grad x0: correct
- r10: Exit, gate-matched pool: valid_prem
- r10: Exit, gate-matched pool: cvf
- r10: Exit, gate-matched pool: correct
- r11: Gumbel + subset loss, 2-GPU DP: online root v vs z
- r11: Gumbel + subset loss, 2-GPU DP: held-out, proof end
- r11: Gumbel + subset loss, 2-GPU DP: held-out, mid proof
- r11: Gumbel + subset loss, 2-GPU DP: valid_prem
- r11: Gumbel + subset loss, 2-GPU DP: cvf
- r11: Gumbel + subset loss, 2-GPU DP: correct
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: online root v vs z
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: held-out, proof end
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: held-out, mid proof
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: valid_prem
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: cvf
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: correct
- r13: Gumbel + subset, positive part only: online root v vs z
- r13: Gumbel + subset, positive part only: held-out, proof end
- r13: Gumbel + subset, positive part only: held-out, mid proof
- r13: Gumbel + subset, positive part only: valid_prem
- r13: Gumbel + subset, positive part only: cvf
- r13: Gumbel + subset, positive part only: correct

- AZ online from e6 (r1, lr 1e-6): iteration (min)
- AZ online from e6 (r1, lr 1e-6): peak GPU GB
- AZ online from e6 (r2, lr 3e-6): iteration (min)
- AZ online from e6 (r2, lr 3e-6): peak GPU GB
- r3: value grad x0.1 into backbone: iteration (min)
- r3: value grad x0.1 into backbone: peak GPU GB
- r4: value head only (x0): iteration (min)
- r4: value head only (x0): peak GPU GB
- r5: completed-Q policy target: iteration (min)
- r5: completed-Q policy target: peak GPU GB
- r6: Exit policy (NTP on found only): iteration (min)
- r6: Exit policy (NTP on found only): peak GPU GB
- r7: Exit, lr 3e-6: iteration (min)
- r7: Exit, lr 3e-6: peak GPU GB
- r8: Exit, 16 sims, 128 expansions: iteration (min)
- r8: Exit, 16 sims, 128 expansions: peak GPU GB
- r9: Exit, lr 3e-6, value grad x0: iteration (min)
- r9: Exit, lr 3e-6, value grad x0: peak GPU GB
- r10: Exit, gate-matched pool: iteration (min)
- r10: Exit, gate-matched pool: peak GPU GB
- r11: Gumbel + subset loss, 2-GPU DP: iteration (min)
- r11: Gumbel + subset loss, 2-GPU DP: peak GPU GB
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: iteration (min)
- r12: Gumbel + subset + Exit (ntp 4), 2-GPU DP: peak GPU GB
- r13: Gumbel + subset, positive part only: iteration (min)
- r13: Gumbel + subset, positive part only: peak GPU GB